

Bharath Velamala

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Technical Skills

Languages: Python, Rust, Java, JavaScript, C++, R, SQL, Shell Scripting
AI/LLM: PyTorch, RAG, Graph RAG, Vector Embeddings, LLM Orchestration, MCP, Google ADK, FastAPI
Data & Cloud: Spark, Kafka, Flink, Apache Beam, Airflow, dbt, BigQuery, Snowflake, Dataflow, Vertex AI, Terraform
Databases: PostgreSQL, NoSQL (Firestore, MongoDB), Redshift, Redis, Spanner (Graph), FHIR Store
Systems: Kubernetes, Docker, CI/CD, gRPC, Raft, Distributed Systems, Concurrency & Multithreading
Certifications: GCP Professional Data Engineer, GCP Associate Cloud Engineer

Professional Experience

Data Engineer II – Centauri Health Solutions, Tempe, AZ **Jul 2025 – Present**

- Architecting **CLARION**, a multi-tenant clinical data platform with **end-to-end ETL ownership**; processed **1M+ weekly records** with schema validation, enforcing dataset quality SLAs and schema contracts for 5+ downstream consuming teams.
- Built a **Map-Reduce inspired inference pipeline** on Dataflow running NER models across **2.5M clinical charts**; hybrid rule-based + LLM classification enriching BigQuery golden datasets.
- Architecting a **Graph RAG system** on **Spanner Graph** over FHIR-native patient records and medical charts; three-way hybrid retrieval (dense vector, BM25, graph traversal) across **10K+ documents** to ground LLM inference in AI-assisted medical coding.

Data Science Intern (Biostatistics) – Mayo Clinic, Rochester, MN **May 2024 – May 2025**

- Co-authored and published **medspaCyV (PMC12374723)**, an open-source clinical NLP tool; custom sentence tokenizer, sectionizer, and NER visualizer converting EHR notes into structured assets.
- Engineered a **reusable, event-driven NLP microservice** on GCP (Cloud Run, Firestore) to translate clinical notes into structured assets; leveraged asynchronous concurrency and optimized service boundaries to reduce pipeline latency by **60%**.

Data Engineer – HSBC, Hyderabad, India **Jan 2023 – Aug 2023**

- Introduced a **multi-tenant data layer** (PostgreSQL/BigQuery) with clustering, partitioning, and tuned SQL; eliminated drift across **20+ critical tables** and partnered with 3+ upstream/downstream teams to achieve **100% schema compliance**.
- Designed and owned **distributed Spark/Dataproc pipelines** for financial analytics on GCP; applied windowed computations and optimized joins to cut processing time by **40%** and cloud costs by **25%**.
- Migrated **large-scale financial pipelines** (2M+ weekly records) to GCP (Airflow, Dataflow, BigQuery), reducing execution time by **35%**; established regression testing suites to safeguard data integrity and ensure reliable production deployments.

Associate Data Engineer – SpringML Inc., Hyderabad, India **Jan 2021 – Dec 2022**

- Engineered a **large-scale Apache Beam pipeline in Java** on Dataflow, orchestrated via Airflow, processing **2M+ weekly records**; tuned autoscaling and sharding for 8-min latency (**50% improvement**) with PII masking for data governance.
- Implemented **fault-tolerant CDC pipelines** on GCP (BigQuery, Dataflow, Airflow); built dbt models with optimized SQL MERGE for delta processing; maintained high platform uptime and accelerated data availability by **35%** for stakeholders.
- Launched a **containerized, event-driven microservice** on Cloud Run to orchestrate Document AI processing via Pub/Sub; handled **100K+ claim uploads** with automated validation workflows, cutting manual verification latency by **38%**.
- Architected **CI/CD + IaC (Terraform, Cloud Build)** across the SDLC; reduced release lead time **40%** with zero manual intervention.
- Deployed a **multi-channel messaging system** (Cloud Run, Dataflow, Pub/Sub); STAR schema in BigQuery boosted stakeholder response rates **30%**.

Projects

Agentic Scratchpad Scheduler | Python, Gemini, NetworkX, Multi-Agent **MLSys 2026 — 2nd Place, Track B**

- Neurosymbolic **multi-agent pipeline** for tensor DAG scheduling on memory-limited hardware; LLM proposes op fusion and traversal structure, deterministic solver enforces constraints and latency; **7.77× aggregate speedup** over baseline.

GitCortex | Rust, tree-sitter, KuzuDB, MCP, gRPC

- Branch-aware **code knowledge graph**: tree-sitter AST indexing (<500ms), embedded KuzuDB, **MCP server** for AI assistants; blast-radius analysis and GitHub Actions CI bot.

WAL Distributed | Rust, Tokio, gRPC/tonic, Raft, Prometheus

- Write-Ahead Log + full Raft consensus**; crash-safe segment engine at **429 MiB/s**, **46ns** encode; leader election, log compaction, InstallSnapshot RPC; actor model, Prometheus metrics.

GCP Dataflow Template MCP Server | Python, MCP, Cloud Run, Apache Beam, Docker

- Built and **deployed an MCP server** (Cloud Run, streamable-HTTP) exposing Dataflow pipeline scaffolding as agent-callable tools; enables AI coding assistants (Claude, Cursor) to create, inspect, and deploy standardized Beam pipelines.

Education

University of Arizona

Aug 2023 – May 2025

Master of Science, Data Science | GPA: 4.0/4.0

Indian Institute of Technology Patna

Aug 2017 – May 2021

Bachelor of Technology, Mechanical Engineering | GPA: 3.5/4.0